

Abhay Tiwari

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Education

Indian Institute of Information Technology, Raichur, Bachelor of Technology in AI and DS Sept 2023 – Expected May 2027

- GPA: 8.05/10.0
- **Coursework:** Data Structures and Algorithms, Artificial Intelligence, Data Science, Computer Networks, Operating Systems, Object-Oriented Programming, Database Management Systems, Digital Image Processing

Experience

Indian Institute of Technology Hyderabad, Research Intern Jan 2026 – Present

- Research Intern under **Dr. R. Maheswaran**, contributing to machine learning for hydrology and Earth observation across multiple projects currently under review/preprint. Detailed experimental results cannot yet be disclosed.
- Developed an LSTM Encoder-based learnable bottleneck for Extreme Precipitation Event modeling (manuscript in preparation).
- Designed Physics-Informed Frequency-Decomposed Transformers for temporal downscaling of GRACE satellite data (manuscript in preparation).
- Engineered a Residual U-Net CVAE for multi-channel geophysical data with custom spatial regularization and physics-informed learning.
- Member of the research group's hackathon team, winning **1st Place** at **Riverathon 2026**.

Projects

Deep Learning Solutions for Biharmonic Equations | *Python, PyTorch, CUDA, PINNs, Deep Ritz Method* Github Repository

- Developed a comparative framework for solving fourth-order **Biharmonic PDEs** using **Physics-Informed Neural Networks (PINNs)** and the **Deep Ritz Method (DRM)**.
- Implemented both strong-form and variational PDE solvers using automatic differentiation and energy minimization.
- Achieved **1.4% relative error** with PINNs while demonstrating a **10× GPU speedup** using DRM.

Reinforcement Learning for Lunar Landing | *PyTorch, Gymnasium, Stable Baselines3* Github Repository

- Developed a PPO-based reinforcement learning agent for LunarLander-v3 using Stable Baselines3 with 16 parallel environments for accelerated and stable learning.

Agentic Legal Assistant (Law-Agentic System) | *LangGraph, LangChain, FastAPI, Firebase, Google Gemini* Github Repository

- Built a multi-role **Agentic AI** legal assistant using **LangGraph** and a **RAG** pipeline over Indian legal codes.
- Deployed with FastAPI and Firebase for complaint filing, case assignment, and legal workflow automation.

Technical Skills

Languages: Python, C++, C, Rust, SQL

Technologies: NumPy, Pandas, PyTorch, LangChain, LangGraph, Physics-Informed Neural Networks (PINNs), Deep Ritz Method, Django, FastAPI

Tools & Platforms: CUDA, Git, SQLite, PostgreSQL, Google Colab, Jupyter Notebook, VS Code

Coding Profile: Leetcode: Solved 250+ Problems (Link), Top 120 Rank - Amazon ML Challenge 25